Structural Early Warning as a Refinement Trigger

A Forced-2D-Turbulence Utility Benchmark and JHTDB Replication Protocol

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Abstract

Previous MAAT fluid benchmarks tested whether structural supports correlate with shock-like steepening and regular-control behaviour. This paper asks a more operational question: can a structural warning functional trigger adaptive refinement earlier than standard vorticity monitors at the same false-alarm rate?

We define a forced two-dimensional incompressible vorticity benchmark and a predeclared refinement oracle based on future high-wavenumber stress. The MAAT warning functional W_{MAAT} combines robustness loss, controlled activity, spectral-tail pressure, and tail-growth pressure. Thresholds are calibrated on non-event windows to a common false-alarm target of 5%, and monitors are compared by detection rate, median lead time, and lead-coverage utility.

In a six-seed forced-2D ensemble with four refinement events, W_{MAAT} detects all events with median lead time 1.275, giving the best lead-coverage utility 1.275. By comparison, maximum vorticity detects only half the events with utility 0.825, RMS vorticity gives 0.5125, and palinstrophy gives 1.000. An explicit high-wavenumber monitor remains close to W_{MAAT} with utility 1.225. Event-bootstrap resampling gives $\Delta U(W_{\mathrm{MAAT}} - \text{high-}k) = 0.050$ with a wide 95% interval $[-0.700, 0.800]$, so the local pilot does not establish a statistically robust advantage over direct spectral-tail monitoring.

A component ablation is equally informative: removing the robustness-loss factor increases utility to 1.400, while tail-only monitoring gives 1.225 and robustness loss alone fails. Thus the current pilot supports structural early warning as an operational refinement trigger relative to simple vorticity monitors, but it also indicates that the present formula is tail/activity dominated and not yet optimized.

The result is therefore narrow but useful: in this local forced-turbulence pilot, W_{MAAT} provides a practical refinement trigger better than simple vorticity monitors, while the margin over direct spectral-tail monitoring is small and statistically unresolved. No Johns Hopkins Turbulence Database data are analysed in this paper; JHTDB is specified as the next external replication route.

1 Purpose

Papers 60 and 66 used fluid diagnostics as structural probes. Paper 67 changes the question from diagnostic correlation to utility:

> *At the same false-alarm rate, does a MAAT structural warning trigger useful adaptive refinement earlier than standard monitors?*

This distinction matters. A diagnostic can correlate with a stress variable without being useful for simulation control. A refinement trigger must make an operational decision before the numerical mesh or sampling strategy would otherwise miss emerging small-scale structure.

The present paper is deliberately staged:

- Stage 1 uses a local forced-2D vorticity benchmark.
- Stage 2 predeclares a JHTDB replication protocol.
- No public DNS data are downloaded or redistributed here.

2 Warning Functional

For each vorticity snapshot $\omega(x, y, t)$, define support coordinates $(H, B, S_{\text{eff}}, V, R_{\text{rob}})$. Here H measures equation-residual consistency, B measures enstrophy balance, S_{eff} measures controlled activity, V measures low-mode spectral coherence, and R_{rob} is the derived robustness closure:

$$R_{\text{resp}} = (HBV)^{1/3}, \qquad R_{\text{rob}} = \min\Big(R_{\text{resp}}, (HBS_{\text{eff}}V)^{1/4}\Big). \tag{1}$$

The warning functional is

$$\boxed{W_{\text{MAAT}} = (1 - R_{\text{rob}})\sqrt{A}\ (1 + 0.7\, P_{\text{tail}})\,(1 + 2\, P_{\text{growth}})\,,} \tag{2}$$

where A is normalized palinstrophy activity, P_{tail} is high-wavenumber spectral pressure, and P_{growth} is a backward finite-difference tail-growth pressure.

This definition is intentionally causal: only the present snapshot and past tail growth are used. Future information is used only to define the evaluation oracle.

3 Benchmark

The local benchmark evolves the two-dimensional vorticity equation

$$\omega_t + u \cdot \nabla\omega = \nu\Delta\omega - \alpha\omega + f(x, y, t), \qquad u = \nabla^{\perp}\psi, \qquad \omega = -\Delta\psi. \tag{3}$$

The forcing is low-mode and intermittent, producing small-scale stress bursts without downloading external DNS data. Six random initial conditions are evolved on a 64^2 pseudo-spectral grid. A spin-up interval $t < 4$ is excluded from event calibration.

Refinement oracle. The target is not blow-up. The target is a practical adaptive-refinement need: future high-wavenumber stress. Operationally, a refinement event occurs when

$$Q_{\text{ref}}(t) = \sqrt{P(t)}\,(1 + 3P_{\text{tail}}(t)) \tag{4}$$

enters its top decile after spin-up. Here $P(t)$ is palinstrophy.

Equal false-alarm comparison. Each monitor $M(t)$ receives its own threshold θ_M, calibrated on non-event windows so that the false-alarm rate is approximately 5%. The primary utility score is

$$U_M = \text{detection rate}_M \times \text{median lead time}_M. \tag{5}$$

This penalizes monitors that have long lead time only on a small subset of events.

4 Results

Monitor	**Detection**	**Median lead**	**Utility**	$\rho_S(M, Q_{\text{ref}})$
W_{MAAT}	1.000	1.275	**1.275**	0.9908
High-k fraction	1.000	1.225	1.225	0.9787
Palinstrophy	0.500	2.000	1.000	0.6190
Max $\lvert\omega\rvert$	0.500	1.650	0.825	0.2611
RMS vorticity	0.500	1.025	0.5125	0.1406

Table 1: Matched-false-alarm refinement-trigger benchmark. The false-alarm target is 5%, with observed false-alarm rate $\simeq 0.0501$ for all monitors.

Comparison	ΔU_{obs}	**Bootstrap mean**	**95% CI**	$P(\Delta U > 0)$
W_{MAAT}−RMS vorticity	0.7625	0.7431	$[-1.1000, 1.9875]$	0.7629
$W_{\text{MAAT}} - \max\lvert\omega\rvert$	0.4500	0.4343	$[-0.3750, 1.5875]$	0.9055
W_{MAAT}−palinstrophy	0.2750	0.2547	$[-1.1750, 1.6500]$	0.6190
W_{MAAT}−high-k	0.0500	0.0914	$[-0.7000, 0.8000]$	0.6243

Table 2: Event-bootstrap uncertainty for utility differences. Resampling is performed over refinement-event onsets. With only four events, the confidence intervals are intentionally interpreted as pilot uncertainty estimates, not as precision statistics.

Ablation monitor	**Detection**	**Median lead**	**Utility**
W_{MAAT} without robustness loss	1.000	1.400	**1.400**
W_{MAAT} without growth factor	1.000	1.300	1.300
W_{MAAT} full predeclared form	1.000	1.275	1.275
Tail-only monitor	1.000	1.225	1.225
Activity-only monitor	0.500	2.000	1.000
W_{MAAT} without tail factor	0.500	2.000	1.000
Growth-only monitor	0.750	0.850	0.6375
Robustness-loss only	0.000	–	0.000

Table 3: Component ablation at the same false-alarm target. The ablation shows that this pilot is primarily tail/activity driven; the robustness-loss factor is not the source of the observed utility and may damp the trigger for this particular refinement oracle.

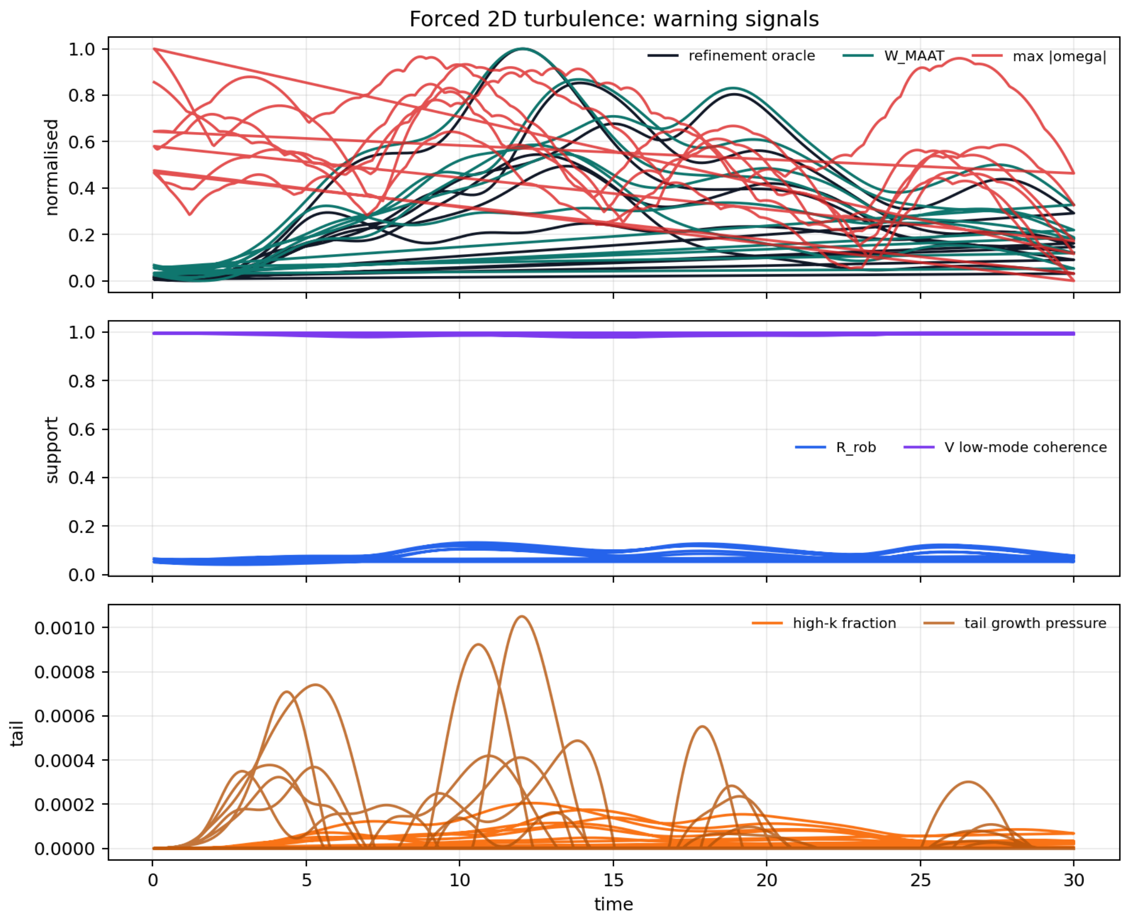

Figure 1: Forced-2D warning time series. The MAAT warning tracks the small-scale refinement oracle more closely than raw vorticity magnitude in the post-spin-up interval.

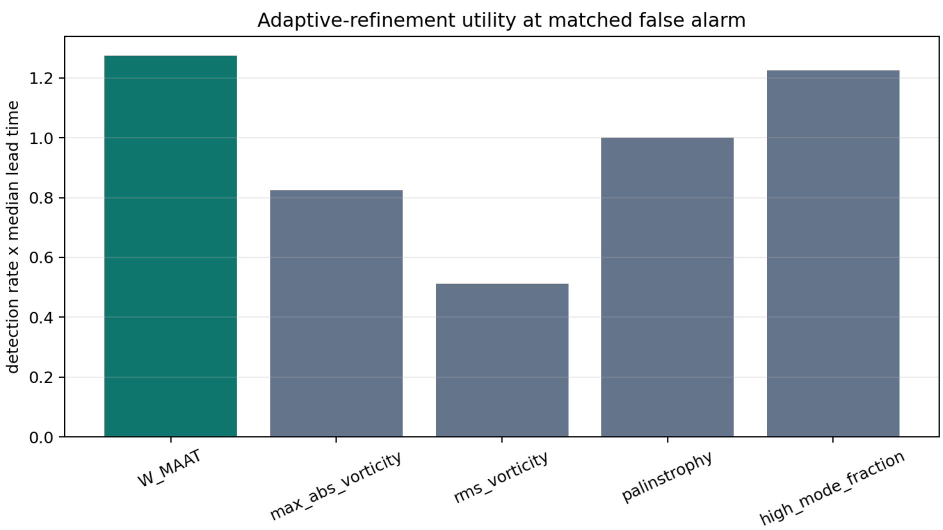

Figure 2: Lead-coverage utility at matched false-alarm rate. W_{MAAT} achieves the highest utility, but only slightly exceeds a direct high-k spectral monitor.

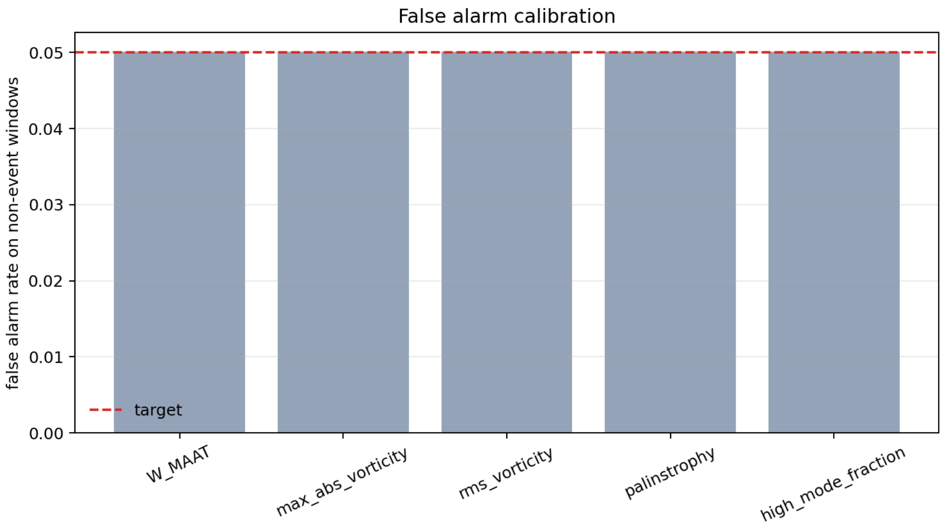

Figure 3: Observed false-alarm rates on non-event windows. All monitors are calibrated to the same 5% target, so the comparison is not driven by different alarm aggressiveness.

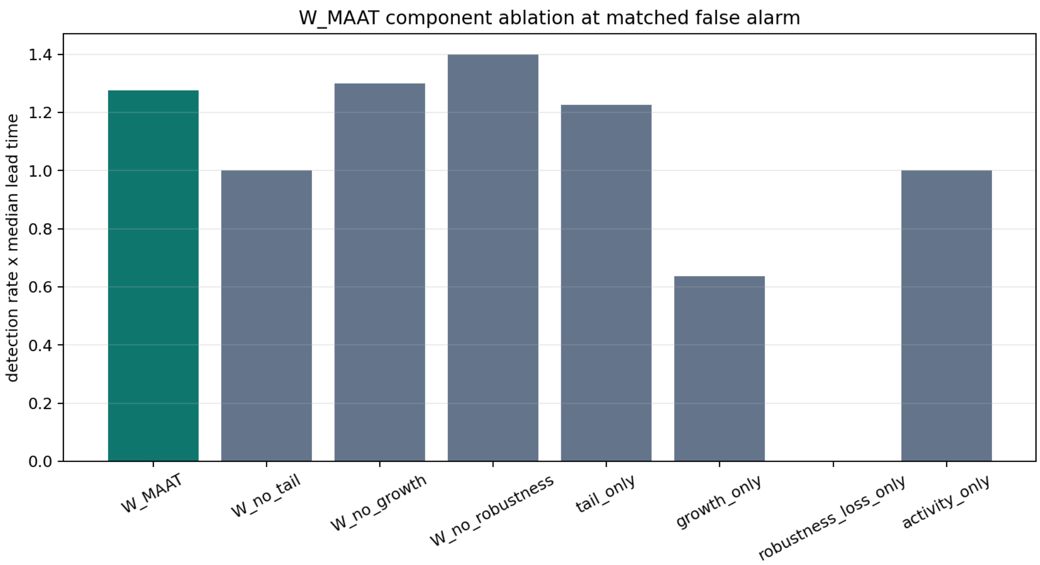

Figure 4: Component ablation of W_{MAAT}. Removing the robustness-loss factor improves utility in this pilot, indicating that the current refinement target is dominated by spectral-tail and activity information rather than by the present robustness closure.

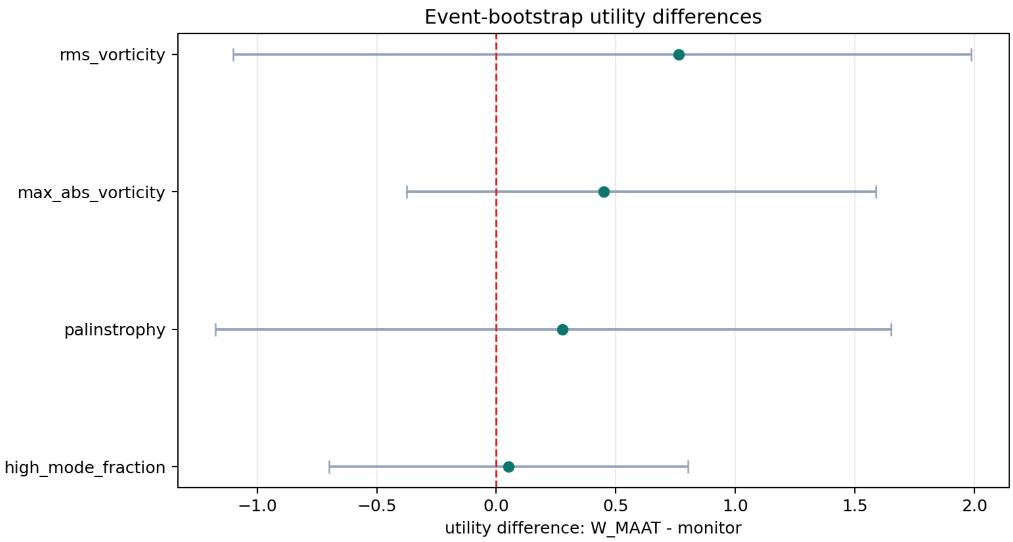

Figure 5: Event-bootstrap utility differences relative to W_{MAAT}. The advantage over high-k monitoring is small and not statistically resolved in the four-event pilot, while the comparison against simple vorticity monitors is more favourable but still limited by the small event count.

5 Interpretation

The useful result is not that W_{MAAT} dominates all possible refinement signals. It does not. The high-k fraction is nearly as good, which is expected because the oracle itself is a high-wavenumber refinement target. The event-bootstrap comparison confirms that this small margin is not yet a robust statistical claim.

The meaningful result is narrower:

> *In this local forced-2D benchmark,* W_{MAAT} *converts structural-support information into a practical refinement trigger that beats simple vorticity monitors at equal false-alarm rate.*

The ablation result is also important. The predeclared W_{MAAT} is not the best possible trigger in the tested family: removing the robustness-loss factor increases utility. For this refinement oracle, spectral-tail and activity information carry most of the useful signal, while the present robustness closure acts more like a damping factor than like an early-warning amplifier. This does not invalidate the trigger result, but it indicates that a future fluid-control version should treat robustness as a gating or regime variable rather than as a multiplicative penalty.

This is the first MAAT fluid benchmark framed as a simulation-control utility test rather than as a diagnostic correlation test. The result remains internal and synthetic, but it defines an operational criterion that can be falsified on public DNS data.

6 JHTDB Replication Protocol

The Johns Hopkins Turbulence Database provides public remote access to large DNS turbulence datasets, including forced isotropic turbulence [1]. Paper 67 does not use those data. The next external version should:

1. select time-resolved JHTDB sub-cubes with velocity or vorticity fields;
2. compute W_{MAAT}, vorticity, palinstrophy, and high-k monitors on identical sub-cubes;
3. define refinement events by future high-wavenumber or gradient-growth stress;

4. calibrate all triggers to the same false-alarm rate;
5. compare detection rate, median lead time, lead-coverage utility, and bootstrap confidence intervals.

No raw JHTDB cutouts should be redistributed unless allowed by current JHTDB terms. Users should cite JHTDB and the relevant dataset publications.

7 Limitations

- The current benchmark is local and synthetic, not a public-DNS result.
- The ensemble contains four refinement events; this is a pilot, not a statistically mature turbulence study.
- The refinement oracle includes high-wavenumber stress, so a direct high-k monitor is a strong and expected baseline.
- The positive utility result is relative to simple vorticity monitors; the margin over spectral-tail monitoring is small and the bootstrap interval crosses zero.
- The component ablation shows that the present robustness-loss factor is not useful for this particular trigger target. This should be treated as a formula-refinement signal, not as a completed fluid-control functional.
- No adaptive mesh solver is actually run. The paper tests trigger timing, not downstream computational savings.

8 Reproducibility

Repository:

https://github.com/Chris4081/structural-selection-principle

Run:

```
cd experiments/structural_early_warning_paper67
python3 structural_early_warning_paper67.py
```

Generated outputs:

- `paper67_forced2d_timeseries.csv`,
- `paper67_trigger_results.csv`,
- `paper67_ablation_results.csv`,
- `paper67_bootstrap_ci.csv`,
- `paper67_event_leads.csv`,
- `paper67_summary.json`,
- `paper67_jhtdb_protocol.json`,
- five PNG figures.

Data and attribution note. The numerical results in this paper are generated from a local synthetic forced-2D turbulence benchmark. No external DNS data are redistributed. JHTDB is referenced only as a future external replication source; use of JHTDB data should follow current JHTDB access terms and cite the relevant JHTDB publications. No endorsement by JHTDB, Johns Hopkins University, IDIES, SciServer, NSF, or the JHTDB dataset authors is implied.

9 Conclusion

Paper 67 provides a first utility-level test of MAAT structural early warning. At matched false-alarm rate in a local forced-2D ensemble, W_{MAAT} detects all four refinement events and achieves the best lead-coverage utility among the tested monitors. This supports the use of structural warning as a refinement trigger relative to simple vorticity monitors.

The refined result is deliberately mixed: W_{MAAT} improves over simple vorticity monitors, but its advantage over a direct high-k monitor is small and not resolved by event-bootstrap uncertainty. The ablation further suggests that the present fluid warning formula should be refined by treating robustness as a regime/gating coordinate rather than as a universal multiplicative warning factor.

The result is therefore not yet an external turbulence validation. The next step is a JHTDB-based replication with public DNS sub-cubes and bootstrap uncertainty estimates.

References

[1] Y. Li, E. Perlman, M. Wan, Y. Yang, C. Meneveau, R. Burns, S. Chen, A. Szalay, and G. Eyink, "A public turbulence database cluster and applications to study Lagrangian evolution of velocity increments in turbulence," arXiv:0804.1703 (2008).

[2] C. Krieg, "Fluid Coherence and Blow-up Diagnostics," preprint (2026).